Bow Valley College

# Course Outline

## ISEC3301

Cryptography
Fall 2021 - Spring 2022

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# ISEC3301 Cryptography

## COURSE DESCRIPTION

In this course, learners use the tools and techniques of cryptography. Focus is on encryption types, digital signatures, data integrity, authentication, protocols, and key management.

| | |
|---|---|
| REQUISITES | None |
| EQUIVALENTS | None |
| CREDITS | 3 |
| HOURS | 45 |
| ELIGIBLE FOR PLAR | Yes |

## COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

**College-Wide Outcomes:**

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

| # | COURSE LEARNING OUTCOME(S) | COLLEGE WIDE OUTCOMES SUPPORTED |
|---|---|---|
| 1 | Set up a software development environment. | 2, 5 |
| 2 | Determine encryption algorithmic needs to implement cryptography and network security. | 2, 5 |
| 3 | Execute symmetric algorithms to secure data transmission. | 2, 5 |
| 4 | Execute hash functions and encrypted hash functions upon a variety of file types, such as text, image, and video. | 2, 3, 5 |
| 5 | Apply public key encryption | 1, 3, 5 |

## COURSE MODULES AND SCHEDULE

**Course schedule subject to change, depending on delivery mode and term of study. For exact dates, please consult the Course Offering Information in Brightspace.*

| WEEK/HOURS | MODULES |
|---|---|
| Week 1 | Module 1: Cryptography and Network Security |
| Week 2 | Module 1: Cryptography and Network Security |
| Week 3 | Module 1: Cryptography and Network Security |
| Week 4 | Module 2: Symmetric Encryption |
| Week 5 | Module 2: Symmetric Encryption |
| Week 6 | Module 2: Symmetric Encryption |
| Week 7 | Module 2: Symmetric Encryption |
| Week 8 | Module 3: Hash Algorithms |
| Week 9 | Reading Week |
| Week 10 | Module 3: Hash Algorithms |
| Week 11 | Module 3: Hash Algorithms |
| Week 12 | Module 3: Hash Algorithms |
| Week 13 | Module 4: Asymmetric Encryption |
| Week 14 | Module 4: Asymmetric Encryption |
| Week 15 | Final Submissions/Exams |

## ASSESSMENT

| COURSE LEARNING OUTCOME(S) | ASSESSMENT | WEIGHT |
|---|---|---|
| 1, 2, 3, 4 | Assignments | 30% |
| 1, 2, 3, 4 | Activities | 30% |
| 1, 2, 3, 4 | Quizzes | 30% |
| 1, 2, 3, 4 | Engagement | 10% |

Important: For details on each assignment and exam, please see the Course Offering Information.

## PERFORMANCE STANDARDS

A minimum grade of D is required to pass this course. However, a program may require a higher grade in this course to progress in the program or to meet specific program completion requirements.

Please consult with the program area or contact the program chair for further details. A minimum Grade Point Average of 2.0 is required for graduation.

## GRADING SCHEME

| **Grade** | **Percentage** | **Grade Point** | **Description** |
|---|---|---|---|
| A+ | 95-100 | 4.0 | Exceptional: superior knowledge of subject matter |
| A | 90-94 | 4.0 | Excellent: outstanding knowledge of subject matter |
| A- | 85-89 | 3.67 | |
| B+ | 80-84 | 3.33 | |
| B | 75-79 | 3.0 | Very Good: knowledge of subject matter generally mastered |
| B- | 70-74 | 2.67 | |
| C+ | 67-69 | 2.33 | |
| C | 64-66 | 2.0 | Satisfactory/Acceptable: knowledge of subject matter adequately mastered |

| | | | |
|---|---|---|---|
| C- | 60-63 | 1.67 | |
| D+ | 57-59 | 1.33 | |
| D | 50-56 | 1.0 | Minimal Pass |
| F | Less than 50 | 0.0 | Fail: an unsatisfactory performance |

## REQUIRED LEARNING RESOURCES

Whitman, M., Mattord, H., Mackey, D., & Green, A. (2013). *Guide to Network Security (*1st ed.). Cengage.

ISBN: 9780840024220

Ciampa, M. (2018). *CompTIA Security+ Guide to Network Security Fundamentals* (6th Ed.). Cengage.

ISBN: 9780357136959 MindTap with ebook access

Additional learning resources may be found in the Course Offering Information or in Brightspace.

## ADDITIONAL INFORMATION

Additional information may be found in the Course Offering Information or in Brightspace.

## ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

## INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Honesty (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

**Turnitin:**

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

**Online Exam Proctoring:**

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.

**Collection of Personal Information:**

This course, including your image and voice, may be recorded and made available to you and other students taking the course section. By attending the class(es) online or in person, you consent to the collection of your personal information. If you do not wish to be recorded, please contact your instructor before starting the course/class to discuss alternative arrangements.

You may use the recordings only for educational purposes and you must not copy, share, or use the recordings for any other purpose without the instructor's express permission.

Your personal information is collected in accordance with section 33(c) of the Freedom of Information and Protection of Privacy Act (Alberta) to deliver academic programming, support learner flexibility, promote universal design for learning principles, and for purposes consistent with the course activities and outcomes. If you have any questions about the collection, disclosure, use, or protection of this information, please contact the College's Access and Privacy Officer at foip@bowvalleycollege.ca.